WEEK 4: VOLUME OF SOLIDS

1. INTRODUCTION

Volume is the amount of **3D space a solid occupies**.
It tells us how much a shape can **hold or contain**.

- Unit: **cubic units (cm^3, m^3)**

Real-Life Applications

- Construction → concrete volume
- Medicine → fluid dosage
- Agriculture → grain storage
- Engineering → fuel tanks
- Packaging → container capacity

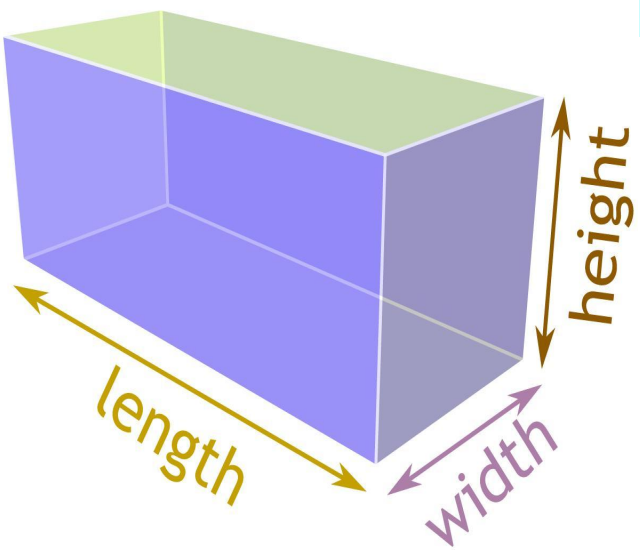
2. CONTENT

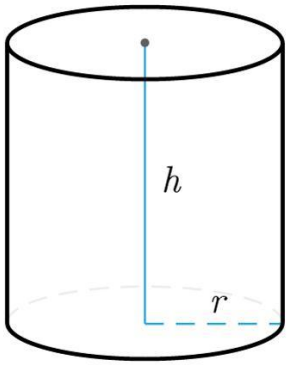
A. PRISMS (INCLUDING CYLINDERS)

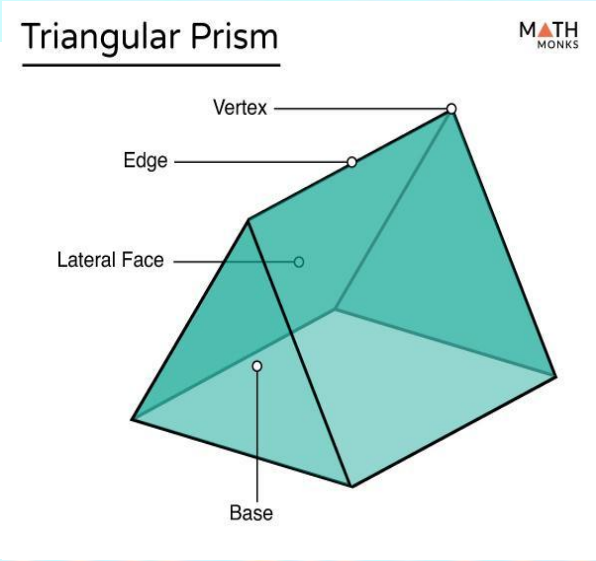
Definition

A prism has a **constant cross-section** throughout its length.

Visual Illustration





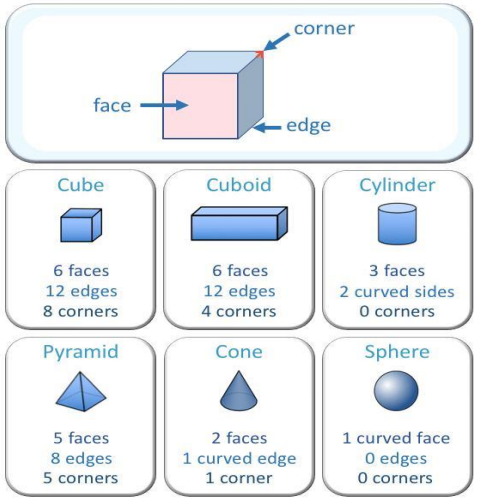


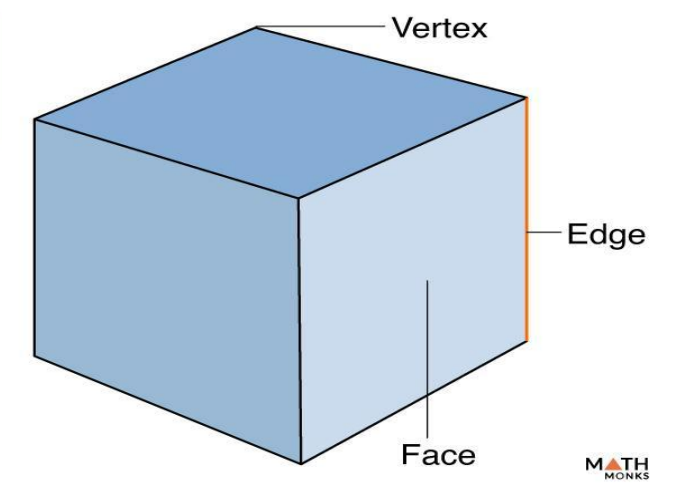
GENERAL FORMULA

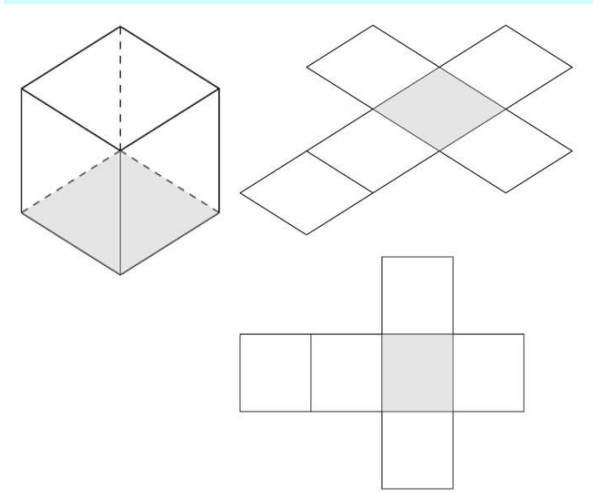
Volume of Prism = Base Area × Height

$V = A \times h$

1. CUBE

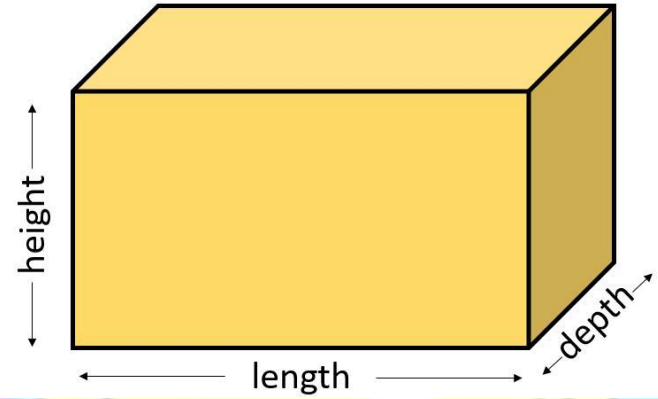


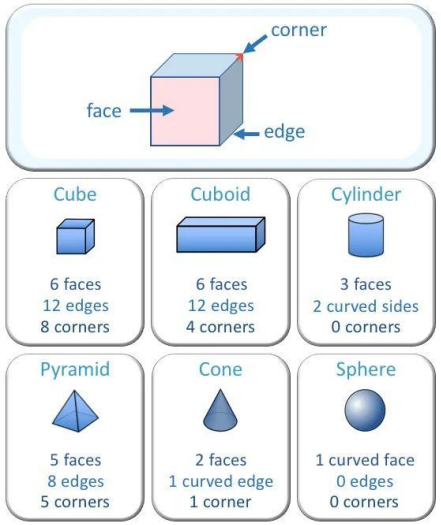


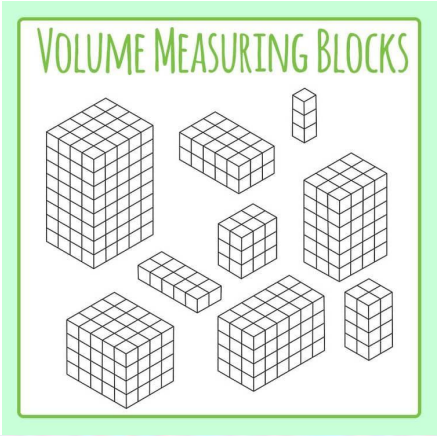


$V = l \times l \times l$
$V = l^3$

2. CUBOID





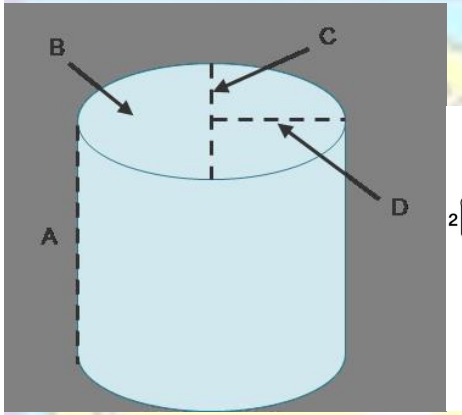


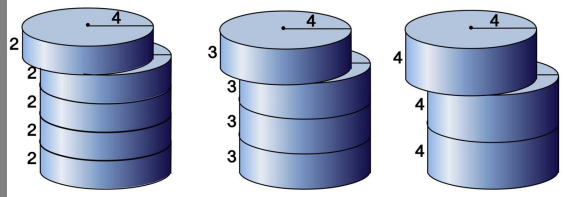
6

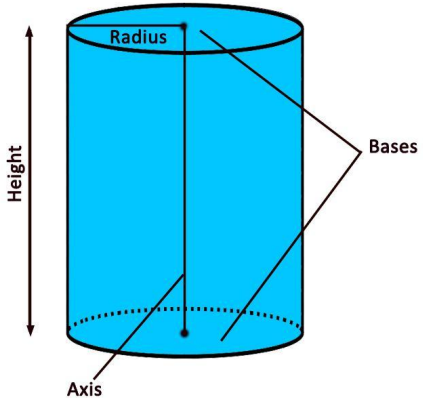
$V = l \times w \times h$

3. CYLINDER

Vedantu
Learn LIVE Online



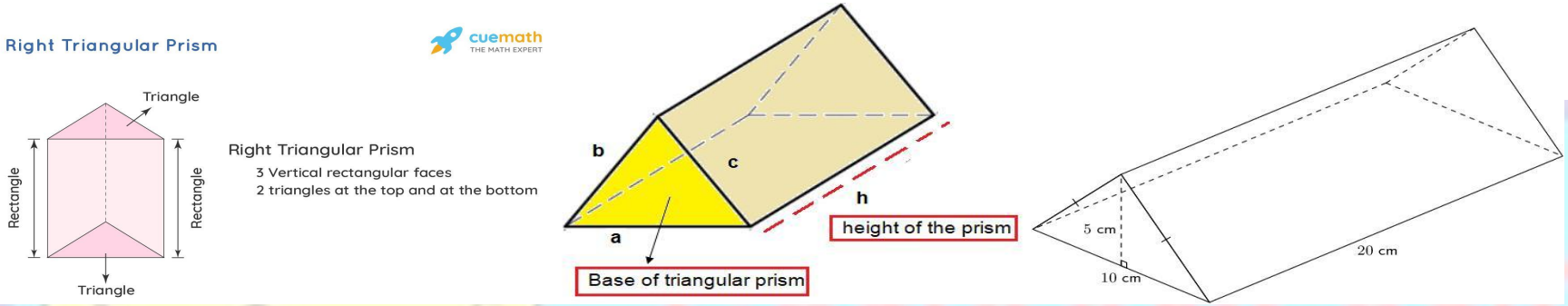




6

$V = \pi r^2 \times h$

4. TRIANGULAR PRISM



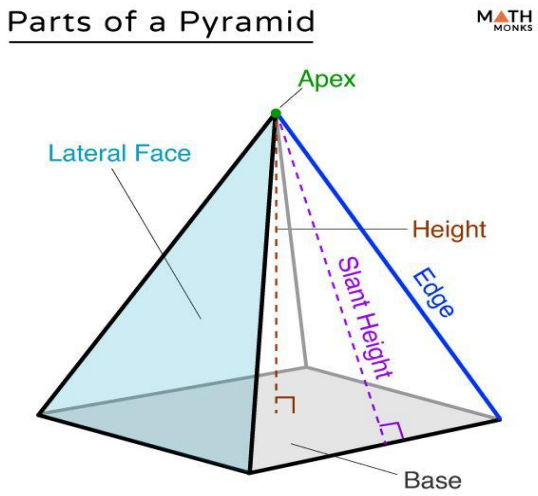
$V = (1/2 \times b \times H) \times L$

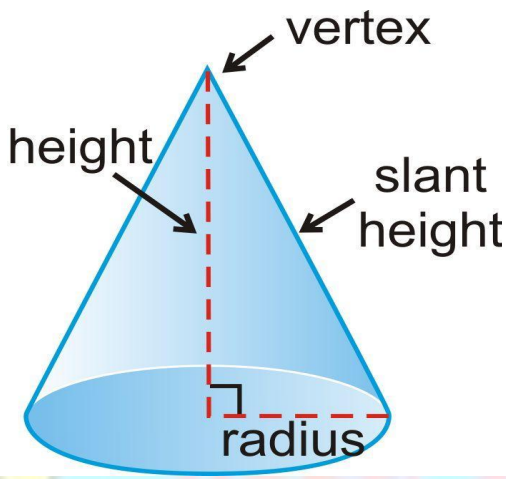
B. PYRAMIDS AND CONES

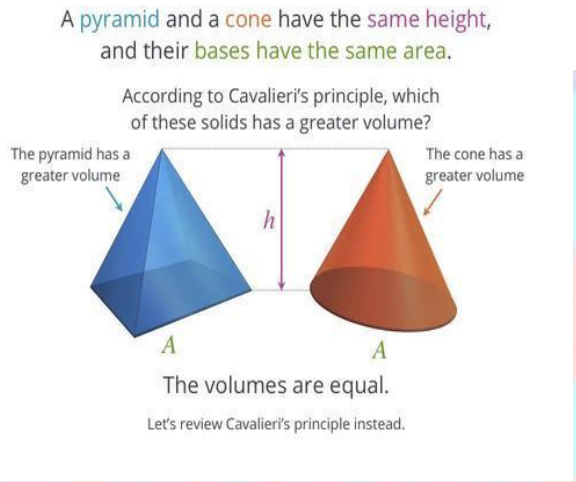
Definition

A pyramid has a base and triangular faces meeting at a point (apex).

Visual Illustration







GENERAL FORMULA

Volume of Pyramid = (1/3) × Base Area × Height

$V = (1/3) \times A \times h$

1. SQUARE PYRAMID

$V = (1/3) \times l^2 \times h$

2. CONE

$V = (1/3) \times \pi r^2 \times h$

IMPORTANT RELATIONSHIP

- Cylinder volume = $\pi r^2 h$
- Cone volume = $1/3\ \pi r^2 h$

Cone is **one-third of a cylinder**

KEY NOTE

✔ Always use **vertical height (NOT slant height)**

3. WORKED EXAMPLES

EXAMPLE 1: CYLINDER

r = 0.7 m, h = 1.5 m

$V = \pi r^2 h$
$= (22/7) \times 0.49 \times 1.5$
$= 2.31\ m^3$

Convert to litres:
$2.31 \times 1000 =$ **2310 L**

✔ Answer: 2310 litres

EXAMPLE 2: CONE (FIND RADIUS)

$V = 110\ cm^3$, h = 21 cm

$V = (1/3)\ \pi r^2 h$

$110 = (1/3) \times (22/7) \times r^2 \times 21$

$110 = 22 \times r^2$

$r^2 = 110 / 22$
$r^2 = 5$

$r = 5$ cm

✔ Answer: 5 cm

EXAMPLE 3: PYRAMID

Base = 10 cm × 10 cm
Slant height = 13 cm

Find height:

$13^2 = h^2 + 5^2$
$169 = h^2 + 25$
$h^2 = 144$
$h = 12$ cm

Volume:

$V = (1/3) \times 100 \times 12$
$= 400$ cm³

✔ Answer: 400 cm³

EXAMPLE 4 (NEW ADDITION): PYRAMID ON CUBOID

CANAANLAND
MODEL GROUP OF SCHOOLS
GOD IS OUR STRENGTH

Question

A pyramid stands on a cuboid with dimensions:

- 4 m × 3 m × 2 m
- Slant edge of pyramid = 5 m

Find total volume.

STEP 1: VOLUME OF CUBOID

$V = 4 \times 3 \times 2$
$= 24\ m^3$

STEP 2: HEIGHT OF PYRAMID

Base = 4 × 3

Half dimensions:

- 2 m and 1.5 m

Distance to corner:

$= \sqrt{(2^2 + 1.5^2)}$
$= \sqrt{(4 + 2.25)}$
$= 2.5$ m

Now:

$5^2 = h^2 + 2.5^2$
$25 = h^2 + 6.25$

$h^2 = 18.75$
$h = 4.33$ m

STEP 3: PYRAMID VOLUME

Base area = 12 m²

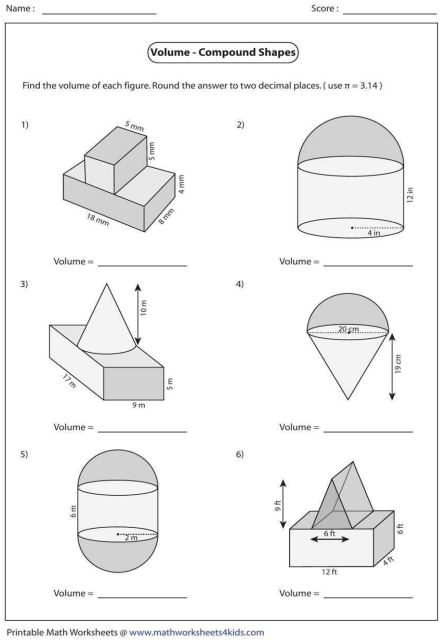
$V = (1/3) \times 12 \times 4.33$
$= 17.32$ m³

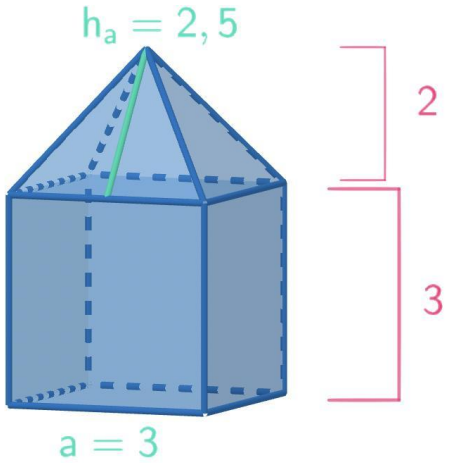
STEP 4: TOTAL VOLUME

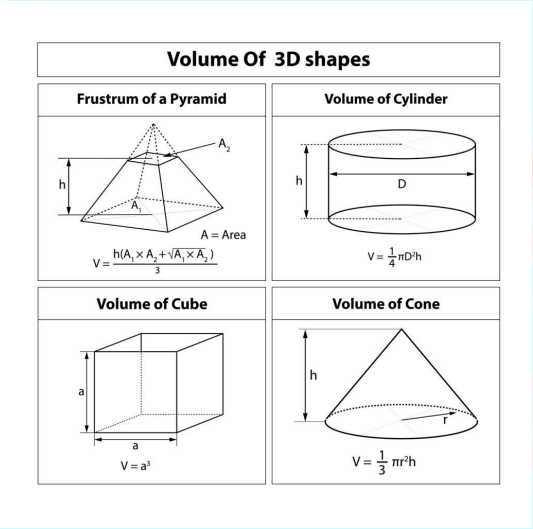
Total = 24 + 17.32
= **41.32 m³**

✔ Final Answer: **41.32 m³**

DIAGRAM







4. REVISION QUESTIONS

1. Formula for prism volume?
2. Formula for pyramid volume?
3. Cone vs cylinder relationship?
4. What height is used in volume of cone?
5. 1 litre = how many cm^3?

5. ASSESSMENT

IN-CLASS

- Cube volume formula
- Cylinder volume formula
- Cone volume formula
- Cube with side 4 cm volume
- Cone vs cylinder comparison

ASSIGNMENT

1. Triangular prism volume
2. Cone volume
3. Cylinder radius problem
4. Square pyramid volume

✅ **SUMMARY**

- Prism = Base × Height
- Pyramid = 1/3 Base × Height
- Cone = 1/3 cylinder
- Always use **vertical height**

CANAANLAND
MODEL GROUP OF SCHOOLS
GOD IS OUR STRENGTH